

AURELIA PALACE

PROJECT SUBMISSION REPORT & TECHNICAL ARCHITECTURE

1. Executive Summary & Introduction

Aurelia Palace is a production-ready, full-stack web application designed for a fictional luxury event venue. The project delivers a refined luxury brand identity, dynamic exploration of venue spaces and services, an interactive enquiry submission pipeline with automated email confirmations, and a secure JWT-authenticated administration portal for lead management.

Engineered to demonstrate enterprise-grade engineering practices, the solution features decoupled frontend/backend services, dual-layer Zod schema validation, managed cloud database persistence (MySQL 8.4 via Prisma ORM), automated transactional emails (Resend API), and strict defense-in-depth API rate-limiting and security headers.

2. Key Links & Project Deliverables

Deliverable / Component	Live Link / Repository Location	Platform
Public Website (Frontend)	https://aurelia-palace-azure.vercel.app	Vercel Edge CDN
Production API (Backend)	https://aurelia-palace.onrender.com	Render Web Service
Source Code Repository	github.com/Jyothi-adithya/AURELIA-PALACE	GitHub Monorepo
Admin Portal	.../admin/login	Vercel / React Router
API Documentation	API_DOCS.md	Markdown Specification
Technical Decisions Note	TECHNICAL_DECISIONS.md	Architecture Note

3. Evaluation & Admin Test Credentials

Admin Login URL	https://aurelia-palace-azure.vercel.app/admin/login
Username / Email	admin@aureliapalace.com
Password	admin123
Admin Capabilities	Review enquiries, filter by status / search, update lead statuses (NEW → CONTACTED → CLOSED), view live conversion dashboard statistics.

4. Technology Stack & Infrastructure

Layer	Technologies & Libraries	Key Purpose
Frontend SPA	React 18, Vite, Tailwind CSS, Framer Motion, React Router v6, React Hook Form, Zod	High-performance responsive UI, cinematic micro-interactions, client-side validation.
Backend REST API	Node.js, Express.js 5, Prisma ORM 5.22, Zod, bcrypt, JWT, Helmet, Morgan	Secure RESTful endpoints, zero-trust schema parsing, stateless token auth, rate limiting.
Database	Managed MySQL 8.4 on Aiven Cloud	Relational ACID persistence for venue content, lead pipeline, and admin users.
Transactional Email	Resend SDK (HTTP REST Integration)	Automated dispatch of luxury-branded HTML confirmation emails.
Cloud Hosting	Vercel (Frontend Edge) + Render (Backend Web Service)	Automated CI/CD deployments connected to GitHub main branch.

5. Major Technical Decisions & Architecture Rationale

1. Decoupled Monorepo Architecture (Vercel + Render)

Rather than utilizing an all-in-one framework, the project enforces clean architectural boundaries with an independent React client and Express API. This enables targeted horizontal scaling, global CDN caching for client assets on Vercel, and specialized Node runtime environment hosting on Render while verifying strict CORS protocols.

2. Dual-Layer Zero-Trust Validation (Zod)

Identical Zod schemas are integrated on both the client (via React Hook Form resolver) and the server (Express controller middleware). This pattern delivers instantaneous user feedback in the UI while ensuring strict data sanitation before any write operation reaches the MySQL database.

3. Relational Persistence with Prisma ORM & MySQL

A relational schema was implemented to model interconnected entities (Events, Spaces, Services, Categories, and Enquiries). Prisma provides compile-time type safety, automated migration management, clean relational queries, and built-in SQL injection prevention.

4. Fault-Tolerant Transactional Email Strategy (Resend)

Confirmation emails are triggered asynchronously following a successful database commit. The database write represents the immutable source of truth. If the email provider returns an error, it is gracefully logged server-side, returning an 'emailSent: false' flag without rolling back the saved enquiry.

5. Stateless JWT Authentication

Administrative access is authenticated through signed JSON Web Tokens transmitted via standard HTTP Authorization Bearer headers. This eliminates server-side session memory, ensuring the API remains completely stateless and robust against horizontal replication.

6. Defense-in-Depth Security & Rate Limiting

The backend employs Helmet for security header hardening, express-rate-limit to protect against brute-force attacks and enquiry form spam, bcrypt for one-way password hashing with salt rounds, and sanitized error boundaries that never leak database credentials or internal stack traces.

6. Verification & Assessment Deliverables Checklist

Item	Requirement	Status
Live Website	Working, responsive Aurelia Palace website deployed to a public URL.	VERIFIED (Vercel)
Source Code	Complete frontend + backend source code in public Git repository.	VERIFIED (GitHub)
Database	Real persistent database implementation with migrations/seed scripts.	VERIFIED (Aiven MySQL)
API Docs	Documented API endpoints and schemas used by the application.	VERIFIED (API_DOCS.md)
README	Setup, architecture, environment variables, and deployment guide.	VERIFIED (README.md)
Demo Credentials	Test administrative credentials provided for evaluation.	VERIFIED (admin123)
Technical Note	Comprehensive documentation explaining major technical decisions.	VERIFIED (TECHNICAL_DECISIONS.md)